

Missingness

Below we summarize and visualize missingness in the dataset.

The number and percentage of missing values for each variable:

```
## # A tibble: 15 x 3
##   variable                                n_miss pct_miss
##   <chr>                                <int>   <num>
## 1 MEMs: Sample interval Decimal Time (mins)    177    47.6
## 2 Booklet: Sample interval Decimal Time (mins)   55    14.8
## 3 DHEA (pg/dl)                                   5     1.34
## 4 Cortisol (nmol/L)                             5     1.34
## 5 DHEA (nmol/L)                                   5     1.34
## 6 SubjectID                                       0      0
## 7 Collection Date                               0      0
## 8 Collection Sample                             0      0
## 9 Booklet: Clock Time                           0      0
## 10 MEMs: Clock Time                             0      0
## 11 Sleep Diary reported wake time               0      0
## 12 Booklet: Sample interval                     0      0
## 13 MEMs: Sample interval                       0      0
## 14 Cortisol (ug/dl)                             0      0
## 15 DAYNUMB                                       0      0
```

Investigating the NAs in MEMs: Sample Interval Decimal Time, it appears that a majority of the NAs correspond to the first time point of Collect Sample = 1.

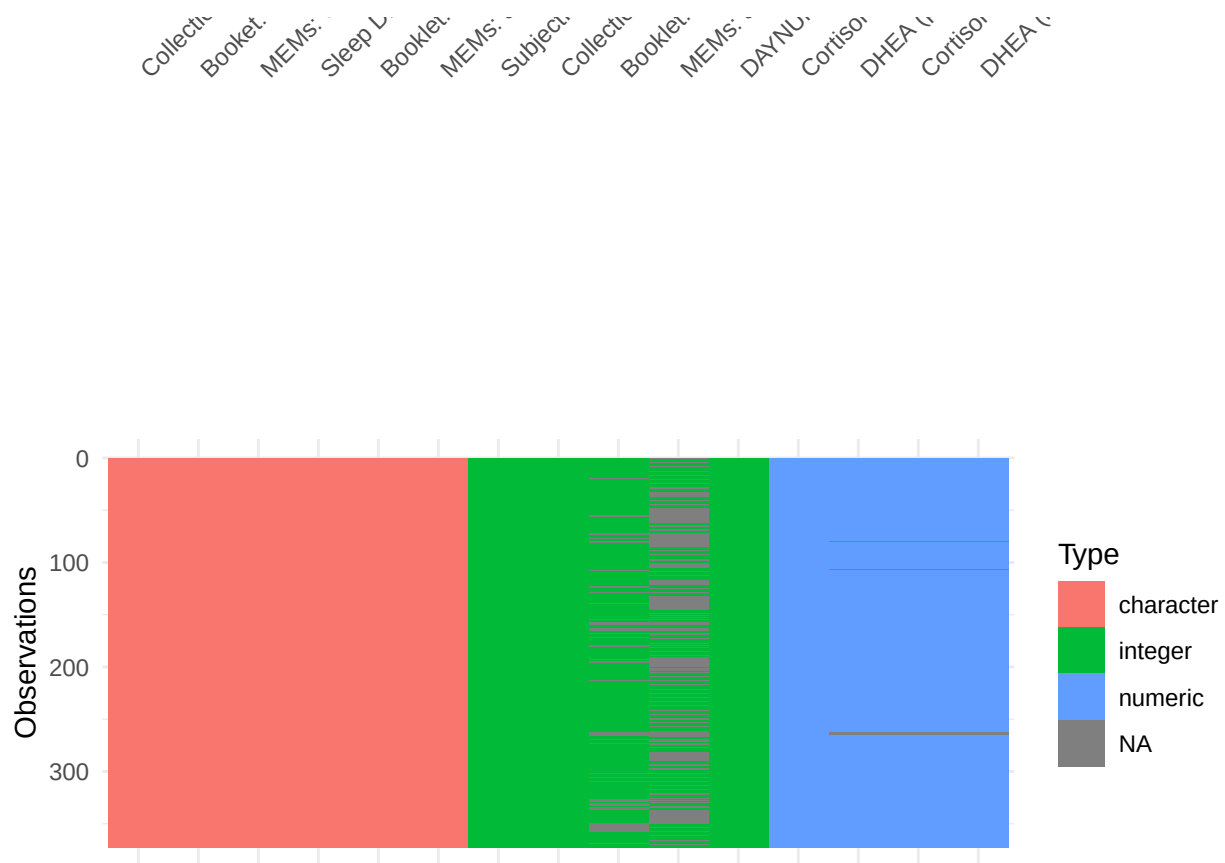
```
## # A tibble: 372 x 3
##   SubjectID `Collection Sample` `MEMs: Sample interval Decimal Time (mins)`
##   <int>          <int>          <int>
## 1     3012           1           NA
## 2     3012           2           43
## 3     3012           3          215
## 4     3012           4          643
## 5     3012           1           NA
## 6     3012           2           36
## 7     3012           3          369
## 8     3012           4          659
## 9     3012           1           NA
## 10    3012           2           34
## # i 362 more rows
```

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```
## # A tibble: 5 x 3
##   n_miss_in_case n_cases pct_cases
##   <int>   <int>   <dbl>
## 1     177     177     0.476
## 2     55      55     0.148
## 3      5       5     0.0134
## 4      5       5     0.0134
## 5      5       5     0.0134
```

##	1	0	174	46.8
##	2	1	162	43.5
##	3	2	31	8.33
##	4	3	1	0.269
##	5	5	4	1.08

Visualizing the missingness:



Descriptive Statistics

A summary of the data - including ranges and data types:

```
##      SubjectID      Collection Date      Collection Sample Booket: Clock Time
## Min.      :3012      Length:372          Min.      :1.00          Length:372
## 1st Qu.:3023      Class :character      1st Qu.:1.75          Class :character
## Median :3032      Mode  :character      Median :2.50          Mode  :character
## Mean    :3033                                Mean    :2.50
## 3rd Qu.:3042                                3rd Qu.:3.25
## Max.    :3061                                Max.    :4.00
##
## MEMs: Clock Time      Sleep Diary reported wake time Booklet: Sample interval
## Length:372          Length:372          Length:372
## Class :character      Class :character          Class :character
## Mode  :character      Mode  :character          Mode  :character
##
##
##
## Booklet: Sample interval Decimal Time (mins) MEMs: Sample interval
## Min.      : 0.0          Length:372
## 1st Qu.: 30.0          Class :character
## Median : 47.0          Mode  :character
## Mean    :243.3
## 3rd Qu.:490.0
## Max.    :955.0
## NA's     :55
## MEMs: Sample interval Decimal Time (mins) Cortisol (ug/dl)      DHEA (pg/dl)
## Min.      : 20.0          Min.      : 0.019      Min.      : 31.00
## 1st Qu.: 39.5          1st Qu.: 0.075      1st Qu.: 89.13
## Median :335.0          Median : 0.153      Median : 176.39
## Mean    :354.7          Mean    : 134.608      Mean    : 282.73
## 3rd Qu.:599.0          3rd Qu.: 0.283      3rd Qu.: 369.86
## Max.    :949.0          Max.    :9999.000      Max.    :1500.00
## NA's     :177          NA's     :5
## Cortisol (nmol/L) DHEA (nmol/L)      DAYNUMB
## Min.      : 0.5242      Min.      :0.1076      Min.      :1.000
## 1st Qu.: 2.0554      1st Qu.:0.3093      1st Qu.:1.000
## Median : 4.1385      Median :0.6121      Median :2.000
## Mean    : 5.9509      Mean    :0.9811      Mean    :2.032
## 3rd Qu.: 7.6562      3rd Qu.:1.2834      3rd Qu.:3.000
## Max.    :89.5571      Max.    :5.2050      Max.    :3.000
## NA's     :5          NA's     :5
```

Question 1

Question 2

Question 3

Cortisol

A summary of Cortisol:

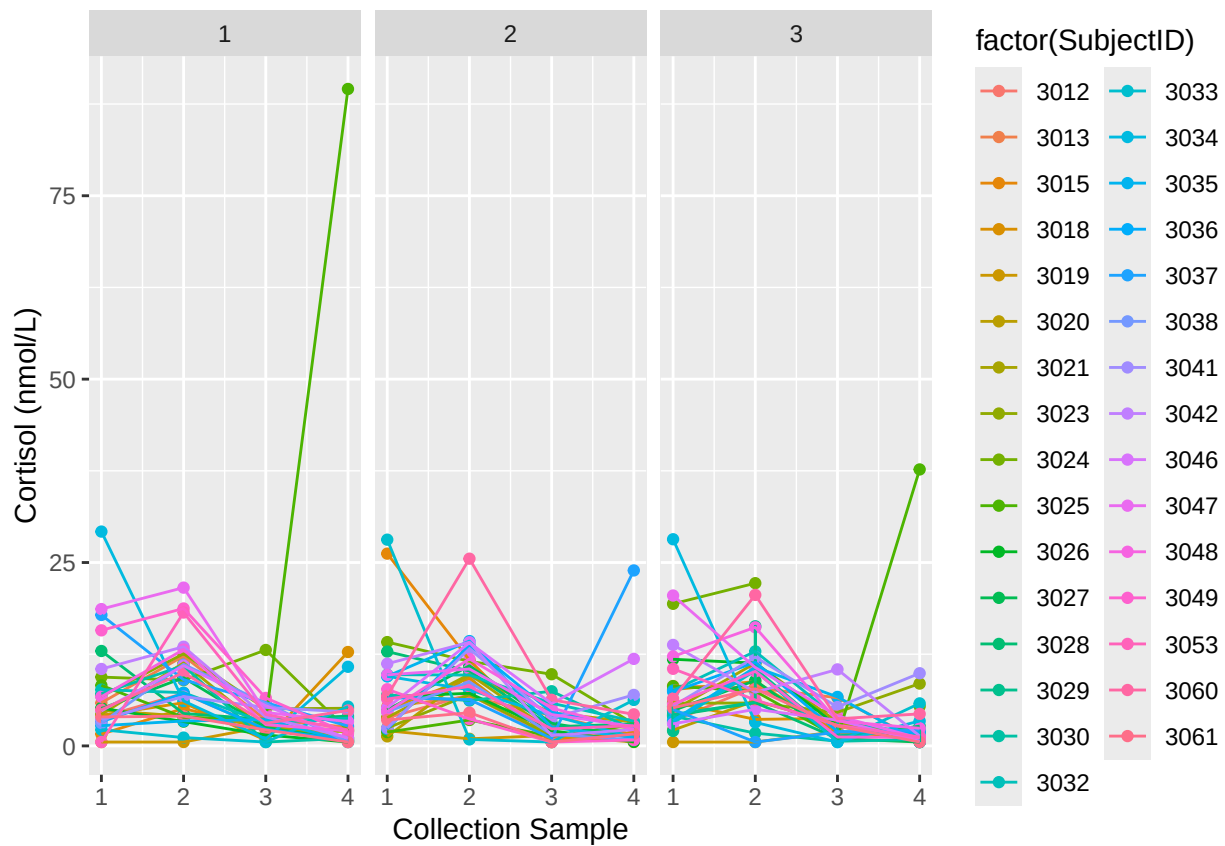
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's  
## 0.5242  2.0554  4.1385  5.9509  7.6562 89.5571      5
```

How many values equal 5.205?

```
## [1] SubjectID      Collection Sample DAYNUMB      Cortisol (nmol/L)  
## <0 rows> (or 0-length row.names)
```

There appears to be one subject that has 4 values of 5.205 for Cortisol (nmol/L).

Visualizing Cortisol over time:



Visualizing a random subset of 5 patients:

Cortisol

A summary of Cortisol:

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
## 0.5242  2.0554  4.1385  5.9509  7.6562 89.5571      5
```

How many measurements are above 26?

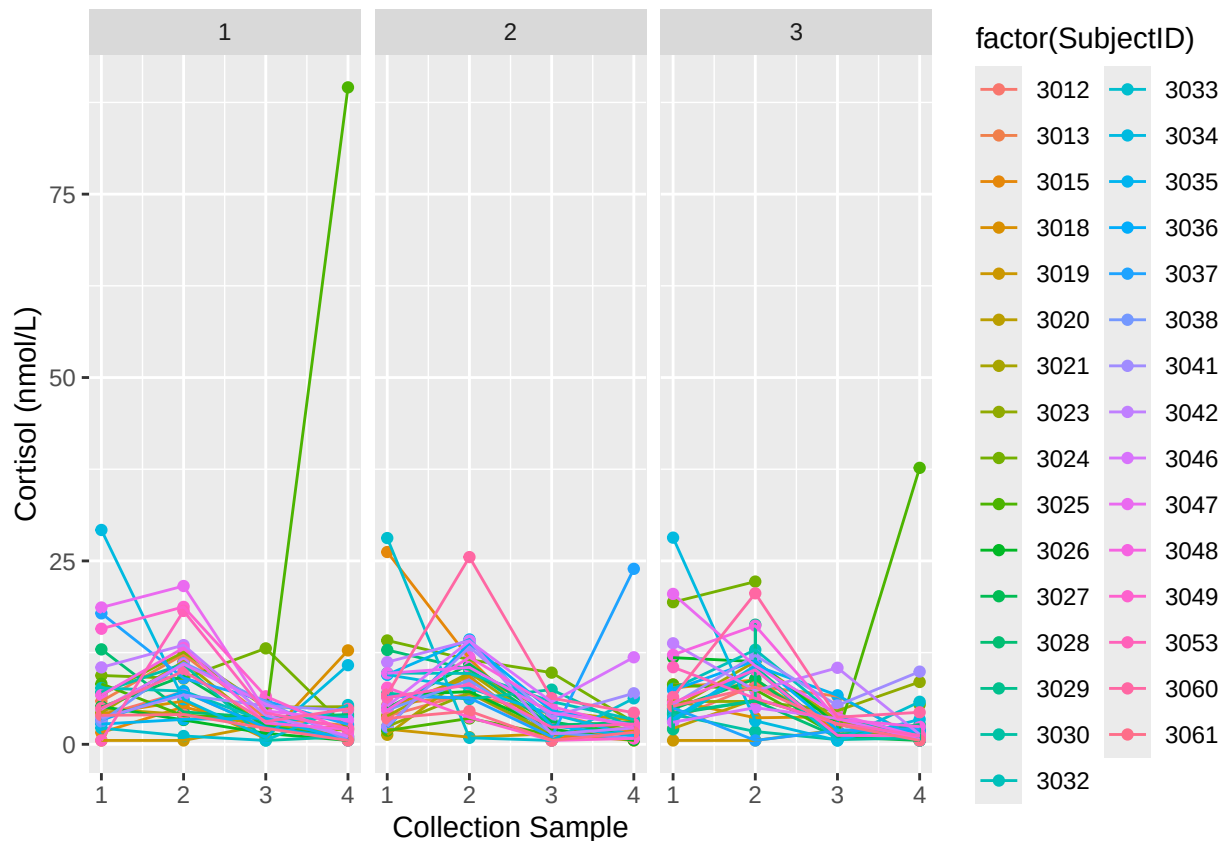
```
## SubjectID Collection Sample DAYNUMB Cortisol (nmol/L)
## 1      3015              1         2      26.21050
## 2      3025              4         1      89.55714
## 3      3025              4         3      37.68794
## 4      3033              1         2      28.11421
## 5      3034              1         1      29.21781
## 6      3034              1         3      28.16939
```

How many values equal exceed 80?

```
## SubjectID Collection Sample DAYNUMB Cortisol (nmol/L)
## 1      3025              4         1      89.55714
```

There appears to be one subject that has a value greater than 80. We will remove this observation for this subject as this is most likely a lab error.

Visualizing Cortisol over time:



Visualizing a random subset of 5 patients: